

Predicting and Explaining Traffic Collision Severity in Toronto
Using Explainable Machine Learning

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Objective

Traffic collisions remain a critical public-safety concern in Toronto, causing hundreds of fatalities and thousands of injuries every year, yet most existing analyses only describe past patterns instead of predicting risk. **This project builds an explainable machine-learning pipeline that predicts collision severity** - Minor (property damage only), Major (one or more injuries), or Fatal (one or more fatalities) - using **677,056 cleaned Toronto collision records (2014–2026)** merged with hourly Government of Canada weather data (Station 31688). Five classifiers are trained and compared, five class-balancing strategies address the severe imbalance (Fatal collisions are only 0.1% of records), and **SHAP** explainability plus **DBSCAN geospatial clustering** are used to interpret model behaviour and descriptively identify spatial concentrations of Fatal collisions that may support further road-safety investigation.

The goal is not just accuracy: a model that is both accurate and interpretable can tell the City of Toronto not only which collisions are likely to be severe, but *why* - and where further investigation may be warranted. Every result below (data, models, explanations, and the map of hotspot zones) is built to answer that question.

Background

Why this matters: Because Minor crashes vastly outnumber Fatal ones, a model that just guesses “Minor” every time looks 86% accurate while missing every life-threatening case - exactly the cases road-safety planners most need flagged.

Data sources: City of Toronto collision records merged, date-hour by date-hour, with Government of Canada hourly weather (temperature, humidity, precipitation) from Station 31688. Invalid GPS records were removed and temporal, weather, and neighbourhood-based features were engineered. Outcome-correlated involvement indicators (pedestrian, cyclist, passenger, motorcycle) identified during the leakage audit were excluded from the final predictive feature set.

Severity class distribution (n = 677,056):

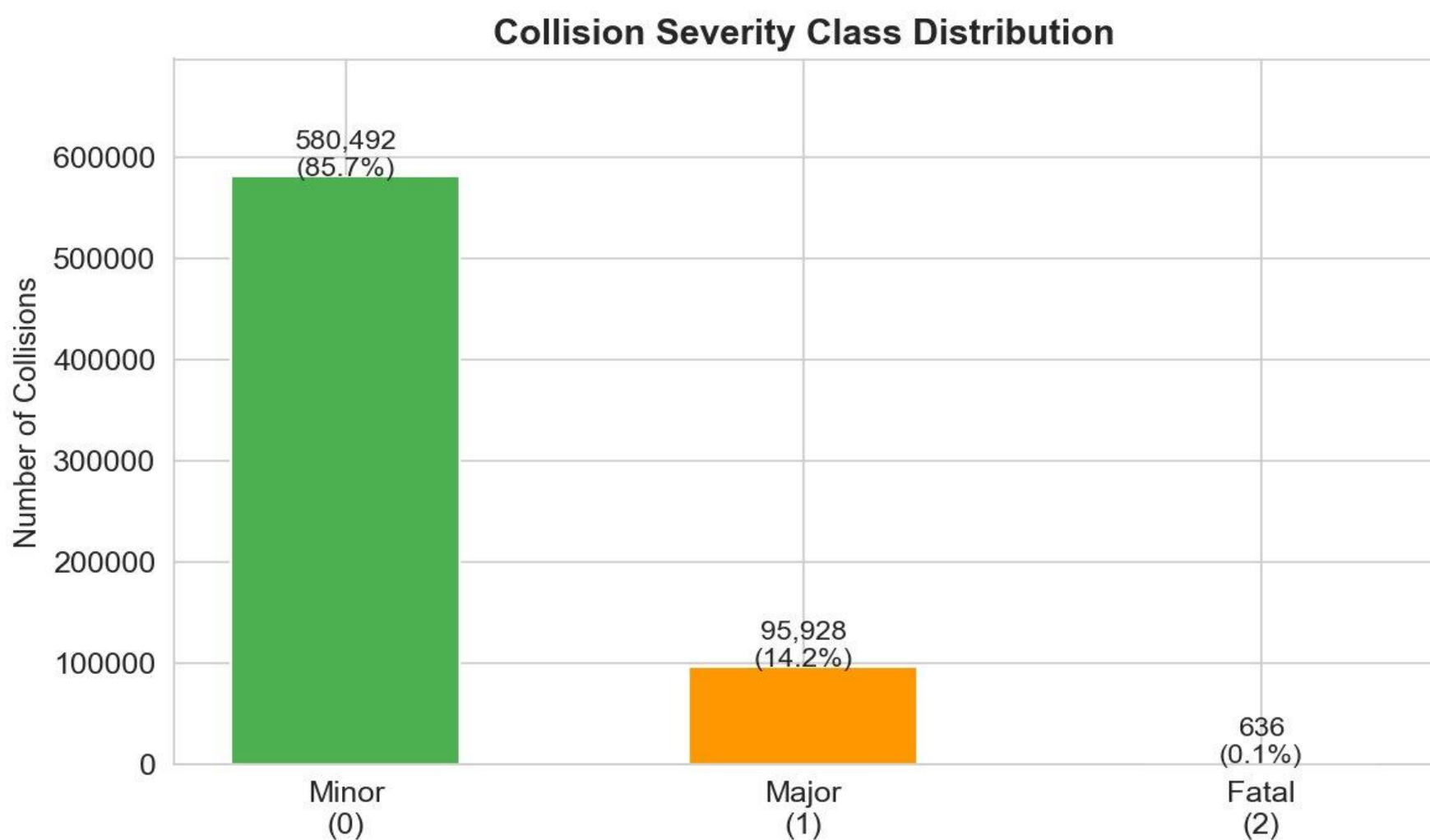


Figure 1. Severity class distribution - Minor 85.7%, Major 14.2%, Fatal 0.1%.

Methodology

Split: 70% train / 15% validation / 15% test, stratified by severity. Headline performance comes from full-dataset 5-fold CV; the test set is used only for SHAP.

Balancing: Five strategies were compared on the validation split — SMOTE-NC, ADASYN, SMOTETomek, no resampling, and random undersampling. For the headline 5-fold CV, SMOTE-NC was applied only within each training fold to prevent resampling leakage.

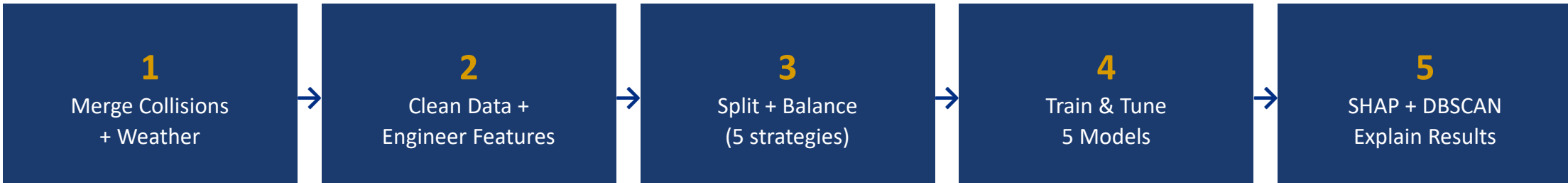
Models: Logistic Regression, Random Forest, Gradient Boosting, XGBoost, and a Neural Network, each evaluated with 5-fold stratified cross-validation.

Tuning: RandomizedSearchCV hyperparameter search on the strongest model, plus decision-threshold tuning to trade precision for Fatal recall.

Explainability: SHAP TreeExplainer was applied to the final XGBoost model using a sample from the held-out test set to examine global feature importance and individual predictions.

Geospatial: DBSCAN clustering (density-based, no need to pre-set a number of clusters) groups the 636 fatal-collision coordinates into 36 dense hotspot zones across the city.

Tools: Python, scikit-learn, XGBoost, imbalanced-learn, SHAP, and DBSCAN - the full pipeline runs from raw CSVs to final explanations reproducibly, notebook by notebook.



Results

Model Performance — 5-Fold Cross-Validation

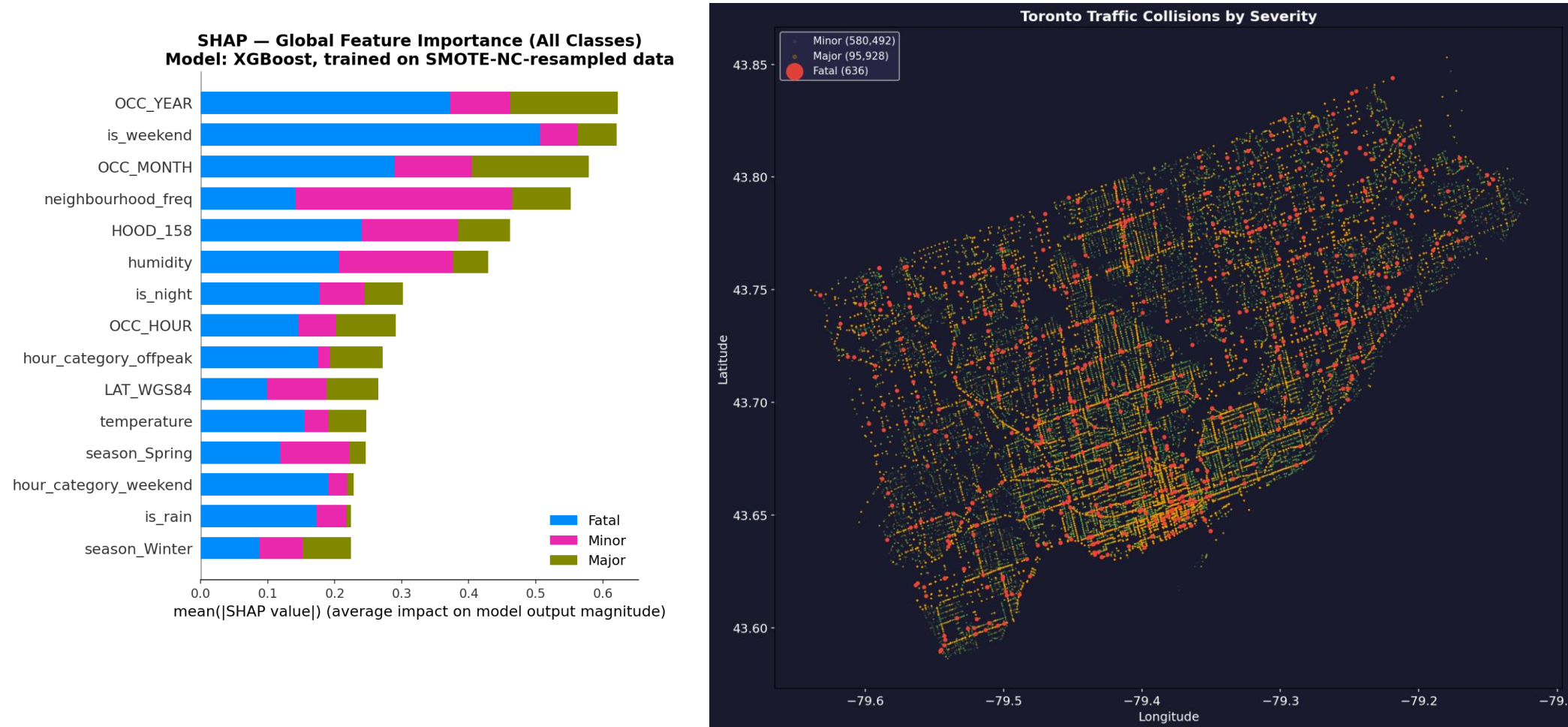
Full dataset (677,056 records) • Corrected pipeline • Five models



The central finding: correcting involvement-flag leakage and a SMOTE-NC encoding error substantially reduced apparent performance from an inflated 0.784 ROC-AUC. Under the corrected, leakage-free, correctly-encoded setup, all five model families show Fatal-class PR-AUC of 0.0010–0.0017, close in absolute terms to a 0.0009 prevalence reference - supporting limited Fatal-class separability under the current predictors and modelling setup, while not ruling out gains from richer predictors or other approaches. XGBoost leads F1 Macro (0.338) and accuracy (81.4%); Random Forest leads Fatal-class PR-AUC (0.0017): neither is an outright winner, and XGBoost still identifies **0 of 95** real Fatal collisions on the validation split.

Balancing and tuning don't fix this either: four of five strategies achieve 0% Fatal recall; only random undersampling recovers meaningful recall (41.1%), at the cost of accuracy collapsing to 37.3%. Lowering the threshold to 0.05 raises Fatal recall to 58.95%, but costs 38.5 accuracy points and precision falls to 0.11% - this study makes *no* deployment recommendation.

SHAP feature importance (left) and DBSCAN Fatal-collision map (right):



is_weekend, OCC_YEAR, OCC_MONTH, neighbourhood frequency, and location lead the SHAP ranking - model *behaviour*, not validated risk factors, given the near-null Fatal PR-AUC. DBSCAN (eps = 300 m) identifies **36 Fatal-collision hotspot clusters** (large red dots), but 499 of 636 Fatal collisions (78.5%) are spatially isolated noise points, not part of any cluster - a descriptive spatial layer, not a comprehensive risk map.

Conclusions

• After correcting involvement-flag leakage and the SMOTE-NC encoding error, Fatal-class discrimination decreased substantially (PR-AUC 0.0010–0.0017, close in absolute terms to the approximate Fatal-prevalence reference level of 0.0009). The corrected results indicate the earlier strong-looking performance (0.784 ROC-AUC) was substantially inflated by these methodological issues.

• XGBoost leads on F1 Macro (0.338) and accuracy (81.4%); Random Forest leads on Fatal-class PR-AUC (0.0017) - neither is an outright winner, and even XGBoost still identifies 0 of 95 real Fatal collisions on the validation split.

• Balancing and threshold tuning do not resolve this: only aggressive undersampling recovers meaningful Fatal recall, and only by sacrificing most of the model's accuracy; hyperparameter tuning can actively hurt Fatal-class detection.

• DBSCAN clustering — unaffected by the leakage and encoding issues above — provides a complementary descriptive spatial analysis: 36 Fatal-collision hotspot clusters were identified across Toronto, including areas around Wexford/Maryvale, West Humber-Clairville, and South Parkdale, while most Fatal collisions remain spatially unclustered (noise).

Limitations:

Fatal collisions remain extremely rare (636 records); a single weather station limits hyperlocal precision; invalid-GPS rate differs by class (16.91% Minor, 12.56% Major, 4.50% Fatal); and the headline CV is not a strictly independent, nested estimate relative to the validation-split selection decisions.

KEY NUMBERS

0.338

XGBoost - F1 Macro

0.0017

Random Forest - Fatal PR-AUC

0 / 95

XGBoost Fatal Recall

36

DBSCAN Hotspot Zones